

Priority	Category	Test	Status	SC Expected Result	Notes
required	vMotion	Live migrate running Linux VM between identical nodes in same cluster			
required	vMotion	Live Migrate running Linux VM between nodes of same CPU brand but different generations			
important	vMotion	Migrate running Linux VM between nodes of different CPU brand			
important	vMotion	Live Migrate running Linux VM between two storage pools in same cluster			
bonus	vMotion	Live migrate running Linux VM between two clusters			
required	HA	Automatic restart of VMs on other nodes/hosts during a node failure			
required	HA	Automatic restart of VMs on another datastore during a storage failure			
required	Performance	Automatic balancing of resources across hosts in the same cluster			
bonus	Performance	VMs cannot be created/Added to the cluster if it violates admission control policies			
required	Deployment	Install custom Linux OS on VMs using a third-party/inhouse PXE service			
required	Deployment	Migrate Virtual machines from vSphere to Spectro			
important	Security	Only VMO admins can create snapshots on any VM on the cluster.			
required	Security	Customized access/privileges to allow a specific group privilege (e.g. snapshots) on specific VMs			
required	Resource MGMT	Ability to group/dedicate a set of resources to a team/project/group to establish a hard cap on resource usage. (e.g. resource pools)			
required	Monitoring	Ability to monitor VM resources utilizations (Memory, CPU, Disk Usage and I/O)			
required	Monitoring	Ability to monitor cluster resources utilizations (Memory, CPU, Disk Usage and I/O)			
required	Monitoring	Ability to monitor resource utilizations on namespace (Or any other resource container we might be using group resources)			
required	Monitoring	Ability to query a specific metric(s)/Properties from a custom collection of VMs (e.g. Power Status). (Note: API endpoint is fine)			
important	Monitoring	Ability to store historical performance data (e.g. one year of performance data)			
important	Monitoring	Ability to store cluster/container performance data.			
required	Log	VM should be able to log events and activities			
important	Log	VM should be able to direct log to central location			
required	Configuration	Ability to create a virtual machine with multiple network interfaces			
bonus	Configuration	Create a VM template from running virtual machine			
required	Configuration	Create a thin provisioned disks for virtual machines			
required	Configuration	Ability to hot add disk(s) to VM			
required	Configuration	Ability to create snapshots of online/offline VMs			
required	Configuration	Configure affinity/anti-affinity rules Between VM-VM and VM-Node(Host)			
required	Automation	Ability to create a VM from an API endpoint			

	Priority	Category
1	required	vMotion
7	required	HA
8	required	HA
10	Required	Performance
11	Bonus	Performance
12	Required	Deployment
13	Required	Deployment
14	important	
18	Required	Resource MGMT
19	Required	Monitoring
20	Required	Monitoring
21	Required	Monitoring
22	Required	Monitoring
23	Important	Monitoring
24	Important	Monitoring
25	Required	Log
26	Important	Log
27	Required	Configuration
28	Important	Configuration
29	Required	Configuration
30	Required	Configuration
31	Required	Configuration
32	Required	Configuration

33	Bonus	Configuration
35	Bonus	Configuration
38	Important	GPU
39	Important	GPU

Test

Live migrate running Linux VM between identical nodes in same

Automatic restart of VMs on other nodes/hosts during a node

Define HA fault domains for clusters

Automatic balancing of resources across hosts in the same

VMs cannot be created/Added to the cluster if it violates

Install custom Linux OS on VMs using a third-party/inhouse PXE

Migrate Virtual machines from vSphere to Spectro

Support for external Vendor supplied images (OVA equivalent)

Ability to group/dedicate a set of resources to a

Ability to monitor VM resources utilizations (Memory, CPU, Disk

Ability to monitor cluster resources utilizations (Memory, CPU,

Ability to monitor resource utilizations on namespace (Or any

Ability to query a specific metric(s)/Properties from a custom

Ability to have/store historical performance data (e.g. one year of

Ability to have/store cluster/container performance data.

VM should be able to log events and activities

VM should be able to direct log to central location

Ability to create a virtual machine with multiple network interfaces

Ability to create VM networks/Portgroups with different security

Create a thin provisioned disks for virtual machines.

Ability to hot add disk(s) to VM

Ability to create snapshots of online/offline VMs

Configure affinity/anti-affinity rules Between VM-VM and VM-

Ability to configure VM CPUs as cores or sockets (for licensing purposes)

vTPM support for KubeVirt VMs (Provider ?)
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vGPU support (GPU Passthrough/partitioning/time slicing) +
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Support for NVIDIA cards and CUDA toolset

[illegible]

Notes

Category	Test	Status	Test
Management Console			
Security	Authenticate users with AzureAD OIDC		
Security	No direct inbound connections from SaaS to managed clusters		
Operations	Consolidated view of entire global fleet in a single pane of glass		
Operations	Consolidated view of regional fleet in a single pane of glass		
Operations	Ability to isolate 'dev', 'non-prod' and 'prod' clusters into different management console versions		
Operations	Show-back/Charge-back at the namespace (or better, app team/group) level		
Kubernetes Cluster Lifecycle Management			
Imported Clusters	Coexist with Rancher agents		
New Clusters	Construct cluster from preexisting Linux VMs and bare-metal machines		
New Clusters	Construct cluster with bare metal (workers) and VMs (control plane) using MaaS		
New Clusters	Global management of cluster pools with similar configs		
New Clusters	Hands-free upgrades		
Disaster Recovery	Automated, periodic etcd snapshots with offsite storage (e.g. S3)		
New Clusters	Automated quarterly OS (kernel, CRI) patching		
Config Mgmt	Per-cluster customization of cluster-wide services, using tag/label selectors		
Config Mgmt	Self-healing of manually modified resources in downstream clusters		
Config Mgmt	Global secrets management for infrastructure services (e.g., a Datadog or S3 API token)		
Kubeconfig	Tenants and admins using kubectl are able to connect directly to the Kubernetes cluster, rather than through a proxy		
New Clusters	Out-of-the-box support for Cilium CNI		
New Clusters	Ability to deploy and manage Kubernetes clusters in 'air gapped' environments with no internet access		
New Clusters	Templated, version-controlled cluster configuration manifests, that update all clusters when modified		
New Clusters	Secrets (and ConfigMap, and...) encryption, with encryption key stored outside etcd node		
New Clusters	Ability to customize/scale CoreDNS as part of cluster configuration template		
New Clusters	Full control over etcd, kube-apiserver, kube-controller, kubelet, kube-proxy configurations as part of cluster configuration template		
Cloud Clusters	Support for imported GKE, EKS, AKS clusters		
Cloud Clusters	Ability to perform full lifecycle management (build, operate, upgrade, retire) GKE, EKS, AKS clusters		
Disaster Recovery	Ability to roll back to any given etcd snapshot in a single operation		
Disaster Recovery	Per-namespace rollback/recovery of deleted or modified resources		
Tenant Management			
Security	All namespaces for a given app team in a cluster have synchronized RBAC		
Security	Automated NetworkPolicy management to prevent app team namespaces from connecting to each other		
Security	Reservation of cluster-wide service namespaces; prevent tenants from taking over critical namespaces		
Security	Tenants can create Secrets that are automatically deployed and synchronized to all of their Namespaces (ideally across all clusters)		
Security	Tenants may not add or remove access to their team/project/group – only admins may do this		
Resource Mgmt	Cluster admin managed ResourceQuotas and LimitRanges for all tenant namespaces		
Resource Mgmt	Out-of-the-box support for Gatekeeper OPA		
Security	Multi-tenant aware security scans and alerts		
Tenant User Experience			
Explorer	Tenants can retrieve Kubeconfig using dashboard UI		
Explorer	Tenants can retrieve Kubeconfig using API		
Explorer	Tenants can create their own namespaces, and may not create namespaces outside their assigned cluster(s)		
Explorer	Tenants can browse their Kubernetes resources in the dashboard UI		
Explorer	Tenants can create new resources using the dashboard UI		
Explorer	Tenants can manage resources across multiple clusters in multiple regions from a single pane of glass		

Priority	Category	Test
required	Deployment	Ability to deploy vClusters into both Palette-managed and imported clusters
important	Deployment	Support for kubernetes versions +/- 2 minor revisions of the parent cluster
required	Deployment	Centralized management of vclusters: monitor, deploy, upgrade, retire
required	Security	vcluster kube-api must only be accessible to cluster admins and to tenants of the vcluster
required	Security	vcluster auth tokens must be issued by Spectro Cloud, e.g. using same Kubeconfig process as for regular clusters, and must follow same tin
required	Security	vcluster tenants must have limited access, e.g. cannot access or mint admin tokens, but can deploy CRDs and ClusterRoles
required	Configuration	admin must be able to fully configure every aspect of the vcluster syncer
important	Deployment	Ability to define templated 't-shirt sizes' for vclusters, small/k3s+sqlite, medium/k3s+etcd, etc.
required	Parent Resources	Workloads in vClusters must be able to utilize parent cluster's ingress controller
required	Parent Resources	Workloads in vClusters must be able to utilize parent cluster's CNI or MetalLB for LoadBalancer Services
required	Parent Resources	Workloads in vClusters must be able to utilize parent cluster's storage classes and use, e.g. Portworx
required	Parent Resources	Workloads in vClusters must be able to utilize parent cluster's observability platform (e.g. Prometheus, Datadog)
required	Parent Resources	Workloads in vClusters must be able to utilize parent cluster's CNI for NetworkPolicy
important	Configuration	vCluster templates should be able to specify managed software deployments inside the vcluster
required	Security	Workloads in vClusters must adhere to security policies of the parent cluster (no privileged pods, no hostPath mounts, etc.)
important	Deployment	vCluster owners/tenants should not be able to interfere with the vcluster control plane pods or configuration
bonus	Deployment	Secure self-service vCluster deployment
important	Deployment	Secure, low latency direct access to vCluster kube-apiserver

Status Spectro expected test result

SC Expected Result Notes

, confirmed upgrade does occur, UI does not reflect new version.

Cluster Groups can limit resources used but vclusters can be created with any size within